

01 / TECHNICAL SPECIFICATION

S-LUT / MOTOMEGA

Functional architecture with F, J and K

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Definition

A finite event-driven system that preserves fixed records, updates a separate reaction-diffusion state, evaluates declared field operators, and returns packed flags or application readouts. A versioned lookup table may replace repeated evaluation for inputs it actually covers.

Event → validate → S-LUT lookup or evaluation → MOTOMEGA T
→ fixed record + updated working state → packed flags / readout

Component	Function
Fixed records / "Deadweights"	Store accepted historical records and reference anchors. Existing records remain unchanged during a working-state update.
Able nodes	Hold numerical U and V values. The F/J/K equations update these values under a declared adjacency and boundary rule.
S-LUT	Maps complete, versioned input keys to precomputed results. A miss requires computation or an explicit unavailable result.
MOTOMEGA T	Defines the transition that keeps locked data fixed while publishing a completed numerical update.
Application readouts	Convert retained state into curves, fields, role assignments or packets. Each readout supplies its own mapping.

Status and source boundary

This is a new formalization of the saved Google AI Mode webpage in this workspace. The webpage is a design dialogue with illustrative code, not a verified implementation.

Equations and bit layout explicitly printed in the webpage are retained where coherent. Record types, transition rules and missing-data behavior below are **selected conventions for this edition**. Undefined geometry, identity and observation mappings remain open; they are not replaced by invented physical laws. [S]

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State and transition contract

$$S = (P, H, D, U, V, R, Q, L, n)$$

Field	Meaning
P	Versioned profile: node count, numeric format, adjacency, field laws, thresholds, timestep, F/J/K rules and capacity limits.
H, D	H is the retained sequence of accepted event records. D is a separate lock mask over numerical nodes; $D[i] = 1$ pins $U[i]$ and $V[i]$. A historical event need not occupy a numerical node.
U, V, R	Two numerical arrays plus declared role assignments. A role has a defined set of permitted local actions; its name does not create sensor capability.
Q, L, n	Bounded FIFO of accepted input events, versioned lookup cache, and increasing commit sequence number.

Input and record

An event carries an event identifier, source identifier, event time, receipt time, payload, units or encoding, and quality information. A committed record adds the profile version, sequence number, result and its classification: observed, derived or predicted. These classifications must survive serialization.

MOTOMEGA T: one complete transition

1. Read one committed snapshot and the FIFO head. Validate its structure, source binding, profile and capacity.
2. Build the full lookup key. Reuse a valid result or evaluate the declared operators for a bounded number of steps.
3. Update only able nodes. Preserve locked values and the existing record prefix.
4. Publish the new record, state and sequence number, and remove the FIFO head, atomically. On failure retain the head and pause for retry or an explicit queue rejection.

$$H' = H \parallel \text{record}; \quad n' = n + 1; \quad Q' = \text{tail}(Q)$$

$$D[i] = 1 \Rightarrow U'[i] = U[i], V'[i] = V[i]$$

Selected execution convention: one ordered writer; readers see completed snapshots. The profile defines how a node becomes locked and its pinned values. Neither a sign test nor an identity label silently changes the lock mask.

Failure and retention

Return `INVALID_INPUT`, `OUT_OF_DOMAIN`, `UNBOUND_OPERATOR`, `CAPACITY` or `NUMERIC_FAILURE` as applicable. An empty or rejected observation is not a zero-valued measurement. Pruning may discard reconstructible cache entries and queued work; deleting authoritative evidence removes the corresponding replay guarantee.

F has two meanings in the dialogue. This edition writes **F_lock** for the historical/palisade constraint and **F** for the numerical feed coefficient. They require an explicit coupling law if intended to be the same operator. [S]

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F / J / K numerical core

An AI reply in the webpage proposes this Gray-Scott-style system. This edition retains it as a selected numerical model. U and V are normalized, dimensionless variables; the 1 in (1 - U) is the reference level. Any physical interpretation requires a scale and calibration. [S]

$$\begin{aligned}\partial U / \partial t &= D_u \Delta U - JUV^2 + F(1 - U) \\ \partial V / \partial t &= D_v \Delta V + JUV^2 - (F + K)V\end{aligned}$$

Parameter	Literal role
$F \geq 0$	Feed/relaxation coefficient driving U toward 1; also participates in V removal.
$J \geq 0$	Multiplier of the nonlinear reaction UV^2 . It changes this reaction rate, not time itself.
$K \geq 0$	Additional V removal coefficient. "Kill" refers to a numerical decay term.
$D_u, D_v \geq 0$	Diffusion coefficients. Their meaning depends on the spatial or graph Laplacian and time units.

Selected discrete binding

$$\begin{aligned}(L_p x)_i &= \sum_j w_{ij}(x_j - x_i) \\ x_i^{\text{next}} &= x_i + \Delta t \cdot \text{RHS}_i \quad \text{when } D[i] = 0\end{aligned}$$

P declares finite, nonnegative symmetric weights w, a time unit, initial U/V arrays, pinned values and Δt . Evaluate both right-hand sides from the same old snapshot. Locked nodes may contribute boundary values to neighboring updates but are not updated. This fixes a gap in the sample code, whose mask is used for initialization without consistently pinning the later evolution.

What J can and cannot do

J may be a fixed scalar or a named function of the current state and accepted event. A dynamic law must specify its inputs, range, update order and cost. $J = 0$ removes the reaction term but leaves diffusion, feed and removal. If V is zero everywhere, the supplied equations do not create nonzero V without a seed or boundary/input term.

The source examples use $D_u = 0.16$, $D_v = 0.08$, $F = 0.0545$ and $K = 0.062$, with several choices of J. These are example values, not a demonstrated stable operating region. Δt and the full reaction-diffusion discretization need a stability bound or analysis; parameter ranges alone do not establish equilibrium.

Two float32 arrays require 8N bytes. Keeping old and next copies requires 16N bytes, before masks, weights, history and caches. Therefore the numerical core is not a one-bit-per-node state representation.

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Field, topology and lookup operators

Operator	Required contract
Signed field / SDF	A scalar field must specify its domain, sign rule and units. A signed distance field additionally needs a metric and boundary. A sign bit retains only a predicate, not distance or geometry.
Double Hadamard product	For same-shaped arrays, choose and record two elementwise products, for example $P = A \odot B$; $Z = P \odot C$. The dialogue also uses $P \odot P^T$: that is a distinct square-array profile. Operand order and the second operand must be explicit.
Hadamard transform	A separate operation $y = Hx$, with $HH^T = mI$ for order m . It is not the elementwise product. The selected Sylvester construction uses $m = 2^k$. [2]
Lowercase phi blend	Set $\phi = (\sqrt{5} - 1)/2$ if the lower golden ratio is intended. Naming ϕ does not specify a blend equation or ensure a distance field. The blend law remains unbound.
Pyramid, circle, cone, apex	Retain these source roles with dimensions, scale, orientation, composition and evaluation law. No exact primitive definition is supplied by the names alone.

Klein and double packing

A possible mathematical domain is the rectangle quotient $(0,v) \sim (1,1-v)$, $(u,0) \sim (u,1)$, for u,v in $[0,1]$. An implementation must state how samples, neighbors and scalar or orientation-dependent values cross those seams. A simple wrap on both array axes gives a torus, not this Klein identification.

“Double Klein” and “double packed” do not specify whether the design uses two fields on one domain, two coupled domains or a product space. This edition retains a two-channel interface with an **unbound topology/coupling law**. A Euclidean inside/outside SDF is not supplied by naming a non-orientable surface.

Log-polar S-LUT

$$\rho = \ln(r/r_0), \quad r > 0; \quad \theta = \text{atan2}(y,x)$$

A log-polar profile declares a reference frame, r_0 , radial/angular limits, bin assignment, angular wrap and origin handling. This uses coordinates internally even if output coordinates are deferred. Quantization and interpolation require stated errors. The LUT key includes profile version, state/event inputs and any time or boundary data that affect the result.

Products of fields are generally not distance fields: even $d(x) = x$ becomes x^2 when squared, losing its sign and distance scaling. A thresholded product is a selected binary feature, not a lossless packing of the fields.

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Binary payload and attached records

The webpage proposes four 16-flag groups in one 64-bit word. Retain this as a **summary payload**. The associated numerical values, identities, anchors and observations live in explicit records, not inside these flags. [S]

Byte offset	Stream bits	Summary role
0-1	0-15	F_lock: fixed-anchor / constraint flags
2-3	16-31	J: enabled phase or role gates
4-5	32-47	Spline: anchor-present / selected flags
6-7	48-63	Observation: recorded-event flags; Doppler only for a declared Doppler input profile

Exact encoding convention

Stream index 0 is the most significant bit of byte 0. Bytes are transmitted in increasing offset order. Let $b[i]$ be exactly 0 or 1; reject other values instead of silently converting arbitrary numbers to truth values. This matches explicit NumPy big-bit-order packing for binary inputs. [1]

$$W = \sum_{i=0..63} b[i] \cdot 2^{63-i}$$
$$b[i] = (W \gg (63-i)) \& 1$$

Thus $b[0] = 1$ alone encodes $0x8000000000000000$; $b[63] = 1$ alone encodes $0x0000000000000001$. These are encoding examples, not execution results.

What must accompany the eight bytes

An envelope identifies schema/profile version, block or node mapping, source, commit sequence, event time and payload length. A referenced attachment supplies the actual anchor positions, coefficients, sample values or records. The identifier-to-record mapping must be retained wherever decoding is required.

Each 16-bit group represents sixteen independent Boolean flags. It does not encode sixteen full spline control points or sixteen continuous observations. An arbitrary 64-flag node block is a different schema and must carry a different schema identifier.

Identity and integrity

A UUID is a 128-bit identifier and therefore cannot fit in this 64-bit word without truncation or an external reference. [3] The source's identifier-seeded random arrays are reproducible synthetic data, not measurements of a person, location or particle field.

A local write-protection flag can prevent ordinary writes through an array interface; it is not cryptographic immutability. If a deployment requires authentication or tamper evidence, its storage and transport profile must supply that mechanism. Pruning a SHA-256 field does not establish equivalent integrity.

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Time, curves and role assignment

History, interpolation and prediction

“Time 0” means the selected event-relative origin t_0 . A timestamp can select a stored result or evaluate a specified model. It cannot determine unobserved position from identity and time alone. Historical lookup returns retained observations; interpolation fills between supplied anchors under a declared rule; extrapolation remains a prediction.

$\text{read}(\text{id}, t) \rightarrow \{\text{recorded}, \text{interpolated}, \text{predicted}, \text{unavailable}\}$
Every result retains its source records and profile version.

Spline readout

A cubic Bézier segment requires four control points $P_0 \dots P_3$ in a declared space and a parameter u in $[0,1]$. A bit may select a point record; it does not supply the point. Catmull-Rom instead interpolates supplied anchors with a declared knot parameterization. [4]

$$B(u) = (1-u)^3 P_0 + 3(1-u)^2 u P_1 + 3(1-u) u^2 P_2 + u^3 P_3$$

For adjacent Bézier segments with equal parameter durations, endpoint agreement and $P_3 - P_2 = Q_1 - Q_0$ give matching first derivatives. Different time durations require derivative scaling. Smooth interpolation is not a proof of a physical trajectory, obstacle avoidance or future motion.

KOEN / J and K-means

Retain KOEN as the source name for permitted role reassignment among able nodes. The profile supplies the role set, eligibility constraints, cost and deterministic tie rule. A role transition cannot modify F_{lock} records or invent data from an absent sensor.

If K-means is used, define feature vectors, distance, cluster count k , initialization, empty-cluster behavior and stopping conditions. Use lowercase k for the cluster count; K remains the reaction removal coefficient. Cluster labels must be mapped to roles explicitly. A clustering step does not provide WAN synchronization.

FIFO, push delivery and shared views

FIFO orders receipt, not necessarily physical event time. Preserve both times; specify duplicate and late-event handling. “Query-free” can mean that producers push events and consumers use an already-maintained view. The view still requires updates.

Multiple local views may read the same committed snapshot. Separate machines need shared inputs or state transfer to agree on new external events. Matching seeds reproduce the same synthetic sequence only under the same generator and execution rules; they do not convey new observations.

A* routing, inverse-square weighting and inverse-log weighting remain application options. Each needs its own graph, domain, cost and admissibility rules. Multiplying weights by $1/\log n$ does not remove tree traversal or establish an optimal route.

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Cost and factual limits

Operation	Defensible cost statement
Pack B flags	$\Theta(B)$ input work and $\text{ceil}(B/8)$ output bytes. One fixed 64-bit mask uses 8 bytes.
Diffusion step	$O(N + E)$ for N nodes and E adjacency entries; $O(N)$ for a bounded-degree stencil. s steps multiply that cost by s .
Elementwise product	$O(M)$ for M array entries. A second product repeats linear work.
Lookup	Direct array access is $O(1)$ only after computing a valid bounded index. Key construction, precomputation, cache misses and output are additional costs.
Tree / clustering	A balanced search tree supports $O(\log N)$ search; an unbalanced tree may take $O(N)$. A direct Lloyd-style K-means pass over N d -dimensional samples costs $O(Nkd)$ per iteration.
History	Storage grows with retained records and payloads. Fixed capacity requires a stop, export or explicit loss/retention policy.

Claims the supplied material does not establish

Source claim	Technical conclusion
Zero energy for a zero bit	Skipping inactive work may reduce activity. Bit value alone does not determine electrical energy; no power measurements are supplied.
Infinite density / "solve infinity"	A 64-bit word distinguishes 2^{64} states. It cannot losslessly represent arbitrary unbounded histories. Topological identifications do not increase finite storage capacity.
Hadamard protocol negation	For the source's symmetric H , $A = HX$ and $B = H^T(-X) = -A$. The layers are negatives, not orthogonal. The reported matrix sum is not an energy measurement.
Exact universal tracking	The source has no calibrated observation model linking its synthetic masks to physical position or personal identity. Deterministic output alone is not evidence.
Perfect security / fixed truth	An immutable record preserves what was recorded, including errors. Read-only arrays, UUIDs and topology do not establish authenticity or cryptographic protection.

The useful efficiency objective is to reduce repeated work: precompute bounded domains, update only affected able nodes when dependencies permit, share immutable records and materialize readouts when needed. Every optimization must preserve the selected numerical and data semantics. No measured speedup, energy saving or universal performance bound is claimed.

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Source trace and remaining bindings

Source basis [S]

Saved webpage: “**americantor set lanl gridder manhattan distance new york social club social small world network graph - Google Search**”. MHTML snapshot dated 23 September 2026, 04:42:36 GMT. Only the active AI Mode conversation is used; unrelated sidebar history is excluded.

The user’s prompts supply the architecture intent. The AI replies supply examples and equations that require assessment. The final photographs are described in the dialogue but do not establish the technical claims. Source SHA-256:

4e39dbca75b4d411892e2155420197b6
107c3d90cbe6050110c3670cf846a941

Source anchor	This edition
“lock the Deadweights ... MOTOMEGA operator T”	Fixed records and pinned nodes, with separate mutable state (pp. 1-3).
“function parameters (F,J,K)”	Retained reaction-diffusion equations; coefficient meanings and limits (p. 3).
“double hadamard ... double Klein bottle”	Separate product, transform and topology contracts; missing bindings explicit (p. 4).
“exact binary bit-mask”	Four 16-flag groups and exact MSB-first encoding (p. 5).
“continuous spline ... F ... dual use”	Anchored readouts and separation of historical F_lock from feed F (pp. 2, 6).
“Koen operator ... K-means mimicry WAN”	Constrained role assignment; clustering and transport treated explicitly (p. 6).

Bindings still needed for a complete implementation

Supply the node/record mapping and lock admission rule; numeric precision and Δt ; the J feedback and KOEN role laws; exact field definitions and second Hadamard operand; double-Klein representation and seams; log-polar bins and thresholds; input calibration, capacities and retention policy. These choices are not determined by the webpage.

Technical references

- [1] NumPy: packbits, bit ordering and byte padding.
- [2] SciPy: hadamard, Sylvester matrix construction.
- [3] IETF RFC 9562: UUID format, 128-bit identifiers.
- [4] Yuksel, Schaefer and Keyser: Parameterization of Catmull-Rom Curves.

Delivery status: document formalization only. Source code was read, not executed or compiled. No implementation, simulation, benchmark or test result is asserted.